

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	16-08-2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Semantic Representation in Customer Support Chatbots.
A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

- Analyze the semantic representations and identify the action-object relationships.
 - Determine which query contains semantic interpretation errors.
 - Evaluate how meaning representation affects chatbot decision accuracy.
 - Recommend improvements to enhance semantic understanding in telecom customer support systems.
2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $Active(x) \rightarrow Producing(x)$
- $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$
- $Maintenance(x) \rightarrow \neg Producing(x)$

Tasks:

- Represent the production data using First-Order Predicate Calculus.
 - Analyze which products are currently available.
 - Infer whether Gear production is affected by maintenance activities.
 - Evaluate the effectiveness of predicate logic in industrial decision-support systems.
3. Word Sense Disambiguation in E-Commerce Search Engines.
An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand

"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

DSA03 – Natural Language Processing

Assessment Tool 1 – Data Interpretation

Q1. Semantic Representation in Customer Support Chatbots

1. Analyze the semantic representations and identify the action–object relationships.

The semantic representations use the form **ACTION(Object, Entity)** to capture the intended meaning of each customer query.

ACTIVATE(Roaming, Customer): The action is *Activate*, and the object is *Roaming*. The customer wants international roaming to be activated.

DEACTIVATE(CallerTune, Customer): The action is *Deactivate*, and the object is *Caller Tune*. The customer wants the caller tune service to be disabled.

QUERY(DataBalance, Customer): The action is *Query*, and the object is *Data Balance*. The customer wants information about the remaining data balance.

Thus, the representations correctly separate the **action** from the **service/entity**, which is important for deciding what operation the chatbot should perform.

2. Determine which query contains a semantic interpretation error.

Q2 contains the semantic interpretation error.

Actual intent: **Deactivate Caller Tune**

Predicted intent: **Activate Caller Tune**

The chatbot correctly identified the entity **Caller Tune**, but incorrectly identified the action. The words *deactivate* and *activate* represent opposite operations. Therefore, the semantic interpretation is incorrect.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation directly affects chatbot decision-making because the system uses the identified action and entity to determine the required operation. If both are correctly represented, the chatbot can provide the appropriate service.

In Q2, the incorrect action could cause the system to **activate a service that the customer wanted to deactivate**. Therefore, even when the entity is correctly detected, an incorrect action representation can result in an incorrect final decision.

4. Recommend improvements.

The chatbot can be improved by:

- Training the intent classifier with more examples of opposite actions such as *activate/deactivate* and *enable/disable*.
- Separately identifying the action and service entity.
- Using contextual information from previous customer messages.

Conclusion: Accurate semantic representation is essential for reliable chatbot decisions. The system should verify both the action and the target service before executing a customer request.

Q2. First-Order Predicate Calculus for Smart Manufacturing

The given production data is:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

The given rules are:

$\text{Active}(x) \rightarrow \text{Producing}(x)$

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

$\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

1. Represent the production data using First-Order Predicate Calculus.

The machine states can be represented as:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Applying the rules:

From **Active(M1) $\rightarrow$ Producing(M1)**

From **Active(M2) $\rightarrow$ Producing(M2)**

From **Maintenance(M3) $\rightarrow \neg$ Producing(M3)**

From **Active(M4) $\rightarrow$ Producing(M4)**

Therefore, M1, M2 and M4 are producing, while M3 is not producing.

2. Analyze which products are currently available.

The rule for product availability is:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

Therefore, a product can be considered available when it is produced by an active machine.

However, the given data does **not specify which machine produces which product**. Therefore, the exact products currently available cannot be determined from the provided data alone.

For example, if:

Produces(M1, ProductA)

and **Active(M1)** is true, then:

Available(ProductA)

can be inferred.

3. Infer whether Gear production is affected by maintenance activities.

The data states:

Maintenance(M3)

and the rule states:

Maintenance(x) $\rightarrow$ $\neg$ Producing(x)

Therefore:

Maintenance(M3) $\rightarrow$ $\neg$ Producing(M3)

If M3 is responsible for producing Gear, then Gear production will be affected because M3 cannot currently produce.

However, the given data does not explicitly state **Produces(M3, Gear)**. Hence, we can conclude that M3's production is affected by maintenance, but we cannot definitively state that Gear production is affected unless the M3–Gear relationship is provided.

4. Evaluate the effectiveness of predicate logic.

Predicate logic is effective for industrial decision-support because it converts machine conditions into formal facts and rules. The system can automatically derive new information from existing machine states.

For example:

Active(M1) $\rightarrow$ Producing(M1)

and

Maintenance(M3) $\rightarrow$ $\neg$ Producing(M3)

allow the system to identify machines that can or cannot produce.

Conclusion: First-Order Predicate Calculus provides a structured method for representing machine states and deriving production decisions. Its effectiveness depends on having complete facts and relationships, such as which machine produces which product.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

The search context provides the clicked results for each ambiguous query.

1. Determine the correct word sense for each query.

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	Clicked result is iPhone Charger
Mouse wireless	Computer Device	Clicked result is Bluetooth Mouse
Java tutorial	Programming Language	Clicked result is Coding Lessons
Python course	Programming Language	Clicked result is Software Development Training

The contextual evidence strongly supports the second sense in every case.

2. Identify the semantic cues used for disambiguation.

The system can use several semantic cues:

- **Apple accessories:** The word *accessories* suggests products, and the clicked iPhone Charger confirms the technology-brand meaning.
- **Mouse wireless:** The word *wireless* strongly indicates a computer device rather than an animal.
- **Java tutorial:** The word *tutorial* and the clicked Coding Lessons indicate the programming language.
- **Python course:** The word *course* and Software Development Training indicate the programming language rather than the snake.

Therefore, the query terms and the user's clicked result provide useful contextual information for WSD.

3. Evaluate the impact of incorrect sense selection.

Incorrect word-sense selection can cause the search engine to return irrelevant results.

For example, interpreting “Apple accessories” as the fruit sense could produce agricultural or food-related results instead of iPhone accessories. Similarly, interpreting “Python course” as the snake sense could produce information unrelated to programming.

This reduces search relevance, user satisfaction and recommendation accuracy. In an e-commerce environment, irrelevant results can also reduce the probability of users finding and purchasing the required products.

4. Suggest an industrial-scale WSD strategy.

An industrial e-commerce system can combine:

- Context-aware language models to understand query meaning.
- Query history and user behaviour.
- Clicked-result information as feedback.
- Product categories and product metadata.
- Search and recommendation history.
- Contextual embeddings to distinguish different meanings of the same word.

The system can combine these signals and select the sense with the highest contextual probability.

Conclusion: WSD is essential for e-commerce search because the same word can have multiple meanings. Combining linguistic context with user behaviour and product information can significantly improve search and recommendation accuracy.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

The healthcare system provides the following syntactic structures:

Doctor prescribed medicine to patient. → Subject–Verb–Object

Patient reported severe headache. → Subject–Verb–Object

Nurse monitored patient continuously. → Subject–Verb–Object

Medicine reduced blood pressure. → Subject–Verb–Object

The generated semantic roles include Doctor–Agent, Medicine–Instrument, Patient–Recipient and Headache–Symptom.

1. Analyze how syntactic structures contribute to semantic role assignment.

Syntax provides the structural relationship between entities and actions. The subject generally identifies the entity performing or causing an action, while objects and other complements provide information about the affected entity, recipient or symptom.

For example:

Doctor prescribed medicine to patient.

Doctor → Subject → **Agent**

prescribed → **Action**

medicine → Object → **Theme**

patient → Recipient → **Recipient**

Similarly:

Patient reported severe headache.

Patient → Subject → **Experiencer/Agent**

reported → **Action**

headache → Object → **Symptom**

Thus, syntactic parsing provides the foundation for assigning semantic roles.

2. Determine whether the assigned semantic roles are appropriate.

Doctor → Agent: Appropriate because the doctor performs the prescribing action.

Patient → Recipient: Appropriate because the patient receives the prescribed medicine.

Headache → Symptom: Appropriate because headache represents the patient's medical symptom.

Medicine → Instrument: This is context-dependent. In the sentence “*Medicine reduced blood pressure,*” medicine can be interpreted as the means or cause through which the reduction occurs. However, in “*Doctor prescribed medicine to patient,*”

Therefore, the assigned roles are mostly appropriate, but **Medicine → Instrument requires contextual interpretation.**

3. Evaluate potential errors from incorrect parsing.

Incorrect syntactic parsing can lead to incorrect semantic roles. In healthcare systems, this can produce serious information-extraction errors.

For example, if the parser incorrectly identifies the patient as the Agent instead of the Recipient, the system may misunderstand who performed the medical action.

Such errors can affect:

Patient records.

Medication information.

Symptom extraction.

Doctor–patient relationships.

Clinical decision-support systems.

Therefore, accurate syntax-driven semantic analysis is particularly important in healthcare applications.

4. Recommend methods to improve syntax-driven semantic analysis.

Healthcare NLP systems can be improved by:

- Using domain-specific medical language models.
- Applying dependency parsing to identify relationships between words.
- Using Semantic Role Labeling (SRL).
- Training models on annotated medical datasets.
- Using medical terminology and ontology resources.

Conclusion

The four scenarios demonstrate that NLP systems must correctly connect language structure, meaning, context and logical relationships. Semantic representation improves chatbot decisions, predicate calculus supports manufacturing inference, WSD improves e-commerce search, and syntax-driven semantic analysis supports accurate healthcare information extraction. Therefore, combining appropriate NLP techniques with contextual and domain-specific information is essential for reliable real-world NLP systems

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____